https://xtremepape.rs/

Question	Answer	Marks
1	**Defining the problem**	
	V is the independent variable and *F* is the dependent variable **or** vary *V* and measure *F*.	**1**
	keep *r* constant	**1**
	Methods of data collection	
	labelled diagram of workable experiment including: • T suspended (clamp or ceiling) • S on top-pan balance vertically below T • T and S labelled and at least one other label.	**1**
	F = difference/change in balance readings when sphere(s) (S) is uncharged and charged.	**1**
	voltmeter across a power supply (with flying lead) **or** read *V* from high voltage power supply/EHT power supply	**1**
	measure *r* with a rule	**1**
	Method of analysis	
	plot a graph of *F* against *V* (allow valid log-log graphs)	**1**
	relationship valid if a straight line passing through the origin is produced	**1**
	α = gradient $\times$ r^2 (consistent with graph)	**1**

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	Additional detail including safety considerations	**Max. 6**
	D1 safety precaution linked to avoiding shock/<u>high</u> voltages, e.g. <u>insulating</u> gloves to hold flying lead/to charge sphere/to avoid electrocution **or** use shrouded leads/ensure that there no bare connections/avoid touching metal parts	
	D2 use of <u>insulator</u> between sphere S and balance or between sphere T and stand	
	D3 discharge sphere(s) by earthing or connecting to the negative of the power supply	
	D4 additional detail to determine *r* e.g. determine radius of each sphere and add distance between spheres **or** measure from top of S to top of T etc.	
	D5 use of calipers/micrometer to measure diameter of spheres (and halve) to determine *r* **or** use of calipers/micrometer to measure diameter of spheres to check that spheres are the same diameter to determine *r* **or** use of fiducial mark and ruler from top of spheres etc.	
	D6 use of Δmg to determine *F*	
	D7 repeat experiment for each value of *V* and average *F*	
	D8 take reading of balance quickly to avoid discharge/keep other charged objects away	
	D9 <u>method</u> to ensure charge on S is constant, e.g. re-charge S periodically/regularly (with initial value of p.d.)/keep S connected to a separate positive terminal	
	D10 avoid draughts to prevent T moving	

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